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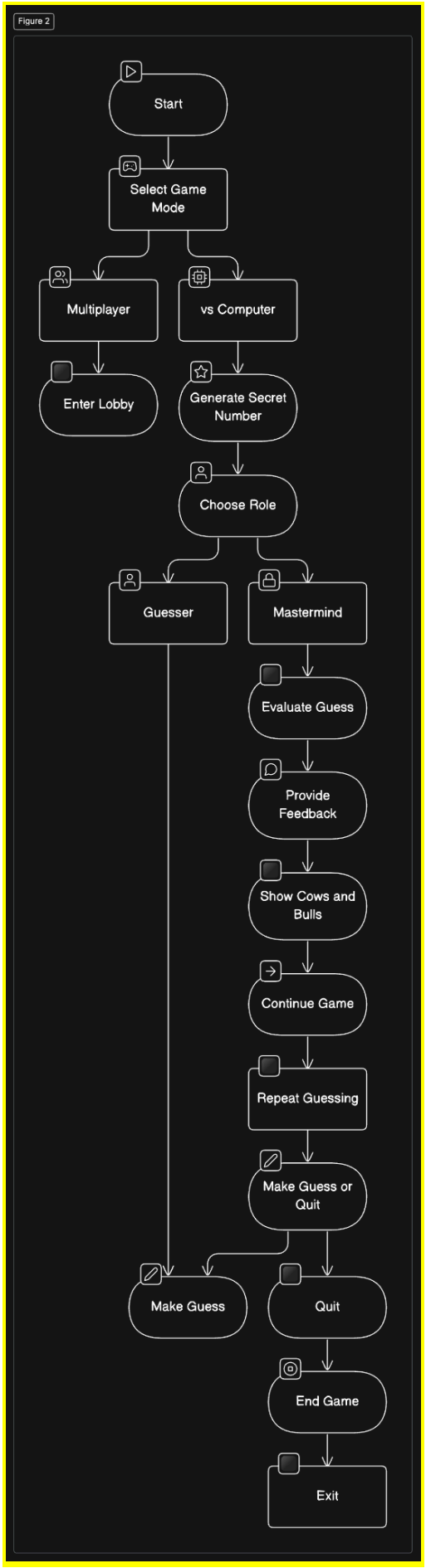
Project Description:

The end product of this project is a web-based implementation of the classic game "Cows and Bulls," also known as "Bulls and Cows" or "Mastermind." The game involves two players: the guesser and the mastermind. The guesser tries to guess a secret code generated by the mastermind within a limited number of attempts. After each guess, the mastermind provides feedback in the form of the number of "cows" (correct digits in the wrong position) and "bulls" (correct digits in the correct position).

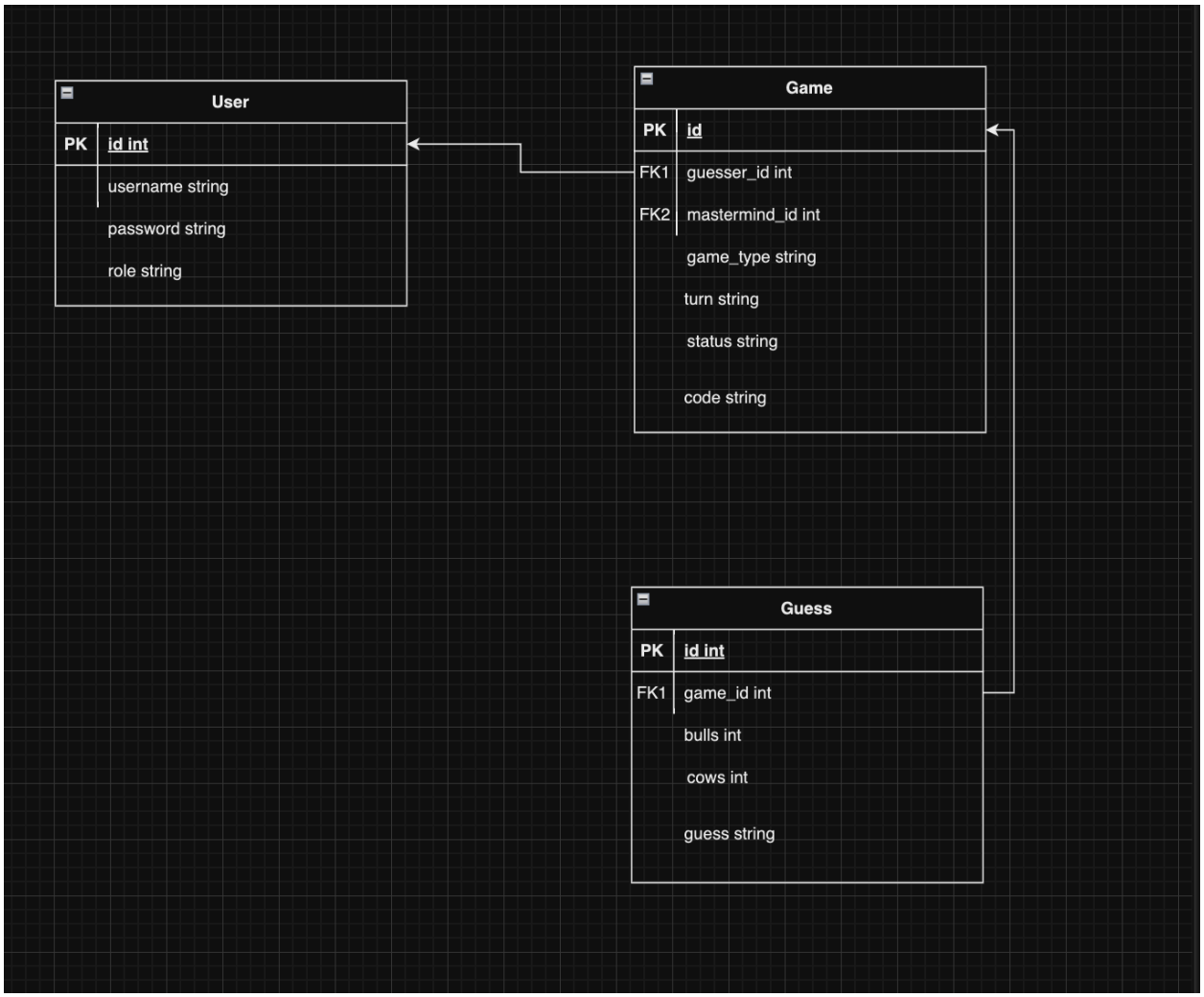
Flowcharts and Diagrams:

Have provided a flowchart illustrating the basic game logic, including player selection, game setup, guess-making, feedback providing, and game completion.

Figure 2


Start
Select Game Mode
Multiplayer
vs Computer
Enter Lobby
Generate Secret Number
Choose Role
Guesser
Mastermind
Evaluate Guess
Provide Feedback
Show Cows and Bulls
Continue Game
Repeat Guessing
Make Guess or Quit
Make Guess
Quit
End Game
Exit

An Entity-Relationship Diagram (ERD) illustrates the database schema, showcasing the relationships between different entities, such as User, Game, and Guess:



Methods Used in Puzzle Solver:

For guess evaluation: Iterate through each digit and compare it with the code.
For the mastermind role: Random code generation and simple guess strategies such as generating random guesses or systematically iterating through all possible combinations.
For the guesser role: Making guesses based on heuristic strategies or Brute Force Search

Market Space and Selling Points:

The application falls within the gaming and entertainment market space. Its selling points include:

Classic and engaging gameplay that appeals to a wide audience.
Multiplayer functionality, allowing users to compete against friends or strangers online.
User-friendly interface and intuitive gameplay mechanics.
Educational value in promoting logical thinking and problem-solving skills.

Functional Specifications:

Product features:

- User authentication: Register, login and save progress functionalities
- Admin account: Account removal
- Game Types: Support for multiplayer, single player
- Player Roles: Option for players to choose between guesser and mastermind roles in multi-player and vs_computer (single) modes.
- Code Generation: Mastermind can set a secret code for the guesser to solve.
- Guess Generation: Guesser can set the code and the machine will try to guess.
- Guessing Mechanism: Guessers can input their guesses, and the system provides feedback based on the code(number of cows and bulls).
- Game Status: Tracking of game status (started, finished, waiting) and turn (guesser or mastermind).

Multiplayer Features : Ability to join multiplayer games, view available games, and play against other online users.

Deployment:

To deploy the Flask project:

- Set up the necessary environment with Python and Flask installed.
- Clone the project repository from the version control system (e.g., GitHub).
- Configure the database connection settings (if applicable).
- Run the Flask application using the flask run command.
- Optionally, deploy the application to a hosting platform like Heroku for public access.

Major Milestones:

Milestone 1 (M1): Basic app setup, authentication, guessing mechanism built
Milestone 2 (M2): Multiplayer support added, allowing users to join and create games.
Milestone 3 (M3): Single player support, code and guess generation
Milestone 4 (M4): Additional features such as game statistics, and leaderboard implemented.
Milestone 5 (M5): Deployment to a production environment and testing with real users.